

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

15th March 2022

Subject SP5–Investment and Finance

Time allowed: 3 Hours 30 Minutes (14.30 - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
- 3. Mark allocations are shown in brackets.*

Q. 1) In a developing economy, the following features are observed:

- disinvestment of public sector - manufacturing as well as services
- establishment of large chains of shopping malls and e-commerce online marts having mergers with multinational companies leading to concentration of pricing power

i) Mention the major economic objectives of a Government's economic policy. (2)

ii) What is a natural monopoly? Give an example. (1)

iii) Discuss key issues the Government should consider in regulating upcoming monopolies. (2)

iv) You have been asked to advise the Government of the developing nation on making changes to the tax system, with the objectives of encouraging entrepreneurial activity and investment to redistribute wealth from the rich to the poor.

Explain the considerations required, in terms of the tax rates for capital gains, investment income and earned income, in order to meet these objectives. (5)

v) In the context of market behaviour at a macro level, when sentiment is LOW, what would you expect to observe with respect to:

- VIX (market volatility index)
- share turnover
- bid-offer spread
- IPO activity and
- the demand for high dividend paying shares (3)

vi) Write short notes on noise traders' impact on market prices. (2)

[15]

Q. 2) You have been hired to work for the Chief Investment Officer of a medium sized insurance company, selling life and pensions products. You are responsible for setting the ALM strategy including consideration of any risks.

i) Define Value at Risk (VaR). (2)

ii) Summarise the stages of an ALM exercise. (3)

iii) Describe how a stochastic model can be used in asset liability modelling to determine the long-term solvency position of a pension fund. (4)

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Q. 3) The manager of a large bond fund is expecting a significant inflow of cash in the near future.

i) Describe how the manager would protect the fund against volatility in bond yields in this situation. (3)

ii) Describe the residual risks that remain with the strategy you have described. (4)

iii) Suggest alternative reasons why the manager might want to hedge the portfolio. (2)

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Q. 4) A fund management firm covering both bonds and equities is analysing the performance of its funds over the last year.

i) An investor is planning to make a switch between the following two bonds:

- Treasury 6% Apr 2026, clean price 122.5, Gross Redemption Yield = 1.35%
- Treasury 5.5% Sep 2025, clean price 118.3, Gross Redemption Yield = 1.41%

Describe the type of switch the investor is undertaking and the investigations to be considered before making this switch. (5)

Several colleagues in the company have been using technical analysis, whereas one of the equity fund managers has not used this technique in the past and is weighing her options.

ii) Outline the possible advantages and disadvantages of technical analysis. (4)

The equity fund manager also noticed that two of her funds has experienced significantly different returns over the past year.

iii) Outline why the performance of two portfolios managed by the same fund manager might differ. (4)

The fund manager is now asked to assess performance of her portfolio against the benchmark. The benchmark rebalances on a quarterly basis and has the following allocation:

30% domestic equities
25% foreign equities
35% bonds
10% cash.

The portfolio's value at the start of the year is INR 5 crores. The fund manager differed from the benchmark by attempting to achieve higher returns via 20% less allocation in bonds, and 10% each more allocated to domestic equities and foreign equities.

At the end of the second quarter, the fund manager rebalanced the portfolio back to the same asset mix as the benchmark. No other cash flows occurred during this period into or out of the fund.

The following information is provided showing the fund value at the end of each quarter (in INR lakhs), prior to any rebalancing, and the benchmark returns.

Value at end of	Quarter 1	Quarter 2	Quarter 3	Quarter 4
Domestic Equities				
Portfolio Value	205	203	170	177
Benchmark Return	8.6%	2.1%	-0.1%	4.4%
Foreign Equities				
Portfolio Value	180	187	137	145
Benchmark Return	5.2%	-0.5%	3.5%	6.6%
Bonds				
Portfolio Value	79	83	191	196
Benchmark Return	5.5%	5.1%	3.2%	2.2%
Cash				
Portfolio Value	50.1	50.2	52.4	52.6
Benchmark Return	0.1%	0.3%	0.2%	0.1%

iv) The fund manager is required to produce the following information:

- a) Calculate the actual quarterly and yearly returns for each asset class in the portfolio, and for the total portfolio. (6)
 - b) Calculate the yearly benchmark returns for each asset class and the quarterly and yearly returns for the total benchmark portfolio. (3)
 - c) State the outperformance or underperformance of the fund versus the benchmark for each quarter and the full year. (1)
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Q. 5) You are an analyst in a stock broking firm responsible for the services sector within India. The growing consensus in the market is that the outlook for the national economy is pessimistic, and the growth rates are likely to be negative for the coming quarter with consumer demand being low.

However, within the company the executives feel that this outlook is likely to change. The company's internal view is that consumers will begin to spend more, and economic growth within the nation will be strong in the coming year or two, although the international outlook continues to be low growth.

Explain how your preferred selection of stocks within the Indian services sector might change based on the company's view.

[4]

Q. 6) A hedge fund launched a new product to its clients (retail investors) that provides variable monthly income. At every month end, after deducting fees from the fund, any amount in excess of capital invested is paid as income for the month to investors.

The fund would charge a monthly

- management fee of 0.2% on AUM (Assets Under Management) and
- performance fee of 20% on fund return in excess of 1% per month net of management fee

The product has a lock-in period of three years post which investors can liquidate their investment anytime by giving one-month notice. The product does not have any entry/exit load but exit load applies during lock-in/notice period.

Initially, a list of good performing stocks is prepared which the fund does not mind holding for long time, if required. The product works in two stages to generate income based on a selected stock from the list.

In stage one, holding sufficient cash, every month sell OTM (Out-of-the-money) put option on the selected stock to collect premium as income until the time the put option is exercised. Upon exercise of the put option, fund holds the stock and stage two begins.

In stage two, every month sell call option (covered call) on the stock that is held to collect premium as income. Strike price of the call option should be equal to the strike of OTM put option that was exercised in stage one. This process of selling covered calls is continued until the time the call option is exercised. Upon exercise of the call option, fund sells the stock and once again stage one begins.

- i) List the reasons why institutional investors might not be interested to invest in hedge funds. (2)
- ii) Discuss possible regulations on the operational (non-investment) aspects of hedge funds. (5)

- iii) Discuss the financial / investment risks associated with the hedge fund's new product. Your answer should include how to monitor and manage these risks. (10)
- iv) Suppose an investor invests one million in the new product of the hedge fund and OTM put option is exercised at the end of first month when stock price was 2% below the strike price (K) of OTM put option. Premium collected at the start of first month is 3.5% of option's notional value (notional value = strike price times the lot size).

Call options expired worthless by the end of second and third month, as stock prices were below K. By the end of fourth month call option was exercised and in fifth month put option expired worthless as stock prices were above K.

The investor's fund value evolves as given in below table.

Month (Stage)	Option Type	Premium (%)	Stock price as %K	FVB	MF	PF	MI	FV
1 (1)	Put	3.5%	98.0%	1,015,000	2,030.00	594.00	12,376.00	1,000,000
2 (2)	Call	1.2%	94.0%	972,000	1,944.00	?	?	?
3 (2)	Call	0.3%	99.0%	1,023,056	2,046.11	?	?	?
4 (2)	Call	2.8%	102.0%	1,038,000	2,076.00	?	?	?
5 (1)	Put	3.2%	105.0%	1,032,000	2,064.00	3,987.20	25,948.80	1,000,000

FVB = Fund value at month end before fee and income; MF = Management fee;
PF = Performance fee; MI = Monthly income paid; FV = Fund value at month end

Calculate:

- Performance fee in 2nd, 3rd and 4th month; and
- Monthly income paid to the investor in 2nd, 3rd and 4th month (5)

v) Comment on the fairness of the performance fee charged

- suggest if any changes required in applying performance fee
- recalculate the performance fee where it is most unfair (4)

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Q. 7) A trader has a view that stock markets would fall in a month and medium term (3 to 7 years) bond yields would fall in six months time. He is considering shorting an equity index future and buying a call option on a Zero Coupon Bond (ZCB).

- i) Describe the role of margin in the context of futures market. (3)
- ii) Calculate variation margins in the first two days for the trader who short an equity index futures at 17,500 level and the index returns were 0.75% and -1.25% for the first two days. Contract (Market lot) size is 50 units. (2)
- iii) State in words the relation between forward price volatility of a bond and forward yield volatility. (1)
- iv) In a developed economy, yield on a Government bond is lower than that of a similar bond in a developing economy. We know that lower yield is associated with higher duration and vice versa.

Does that mean the Government bond in developed economy with lower yield has more interest rate risk? Justify your answer. (3)

- v) A six-month European call option with strike price of CU 80 is available on a 5-year ZCB with face value CU 100. Spot yields are equal to 5% pa compounded continuously for all terms up to six months and forward yield volatility is 20%. (*Note: CU = Currency unit*)

Calculate the price of the six-month call option.

(5)
[14]
